Computer Science & IT



Database Management System

Transactions & concurrency control

Lecture No. 13



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Recap of Previous Lecture



- Topic: Strict 2PL
- Topic: Conservative 2PL
- Topic: Rigorous 2PL
- Topic: Time stamp
- Topic: Read time stamp & Write time stamp

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Topics to be Covered



- Topic: Basic time stamp ordering protocol
- Topic: Time stamp ordering protocol with Thomas write
- Topic: Deadlock prevention using time stamp in lock based protocol
- Topic: Wait-Die protocol
- Topic: Wound-Wait protocol

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Time Stamp Ordering Protocols



Topic : Time stamp ordering protocols

* There are two different versions of time stamp ordering protocol
  1. Basic time stamp ordering protocol
  2. Time stamp ordering protocol with Thomas Write Rule

* Time Stamp :-

Time Stamp is a unique value assigned by DBMS to each transaction in ascending order.

* Let T_1 and T_2 are two transactions in the system, such that

Time Stamp of T_1 < Time Stamp of T_2
$TS(T_1)$ $TS(T_2)$

then, T_1 is the old transaction
&
T_2 is the younger transaction

- Read time stamp of data item 'A'

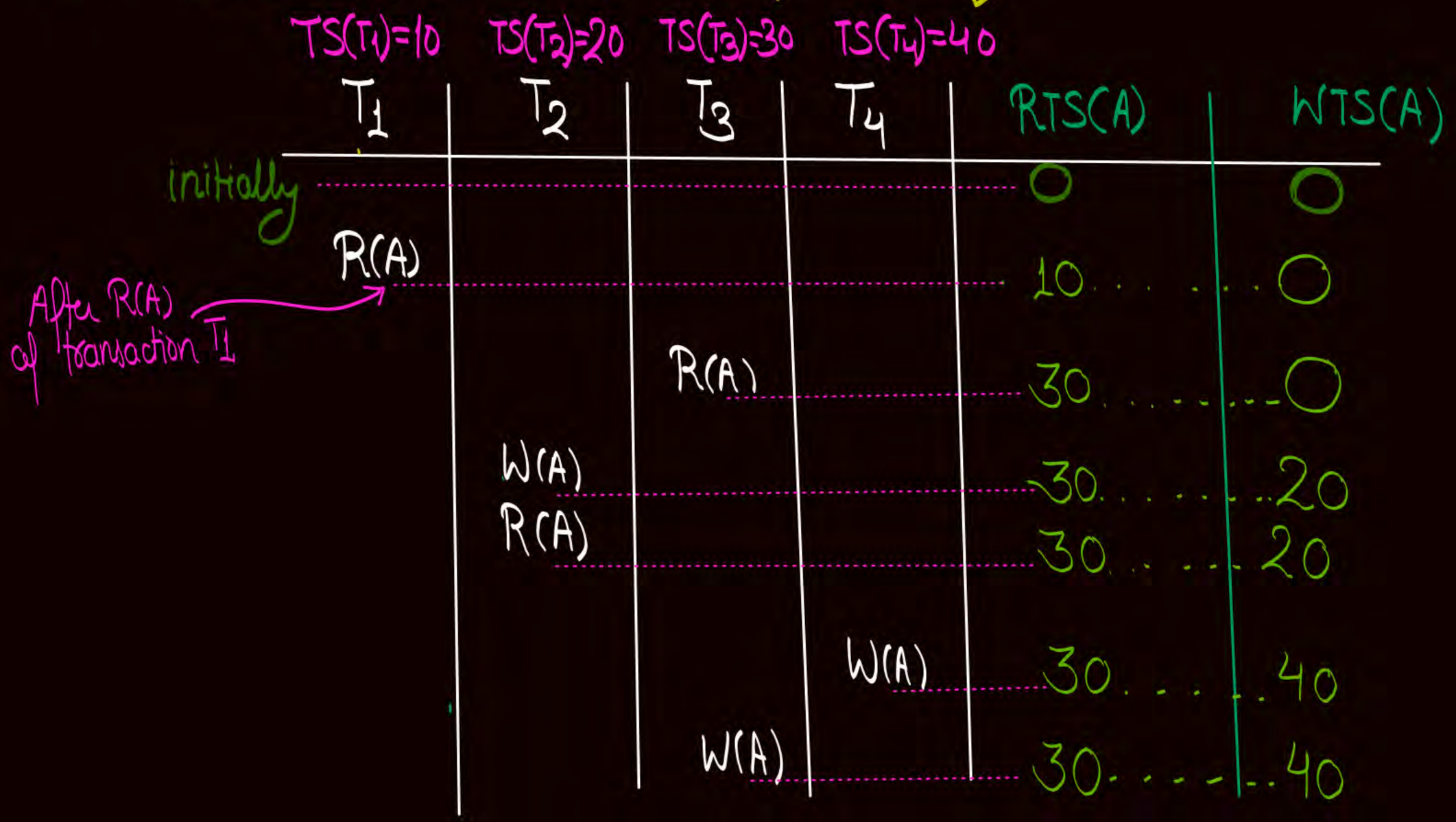
  RTS(A): It is the highest time stamp value among the time stamps of the transactions that has Performed Read(A) opn successfully.

- Write time stamp of dataitem 'A'

  WTS(A): It is the highest time stamp value among the time stamps of the transactions that has Performed Write(A) opn successfully.

- Initially, RTS(A) = 0, WTS(A) = 0 } for all dataitems, not only for 'A'.

eg: RTS(A) & WTS(A) :-


TS(T1)=10
TS(T2)=20
TS(T3)=30
TS(T4)=40
T1
T2
T3
T4
RTS(A)
WTS(A)
initially
0
0
R(A)
10
0
After R(A) of transaction T1
R(A)
30
0
W(A)
30
20
R(A)
30
20
W(A)
30
40
W(A)
30
40



Topic : Basic Time stamp ordering protocol (B.T.S.O.P.)

* A schedule is allowed to execute using B.T.S.O.P. if and only if schedule is a Conflict serializable schedule and Conflict equivalent serial schedule is based on time stamp ordering of the transaction

Eg Consider the schedule 'S' with time stamp ordering as specified along with transactions

S

T_1	T_2	T_3
$TS(T_1)=20$	$TS(T_2)=10$	$TS(T_3)=30$

We can observe that the time stamp ordering is

$TS(T_2) < TS(T_1) < TS(T_3)$

i.e., $T_2 \rightarrow T_1 \rightarrow T_3$

Schedule 'S' will be allowed to execute using B.T.S.O.P. if and only if 'S' is C.S.S. and Conflict equivalent serial schedule is $T_2 \rightarrow T_1 \rightarrow T_3$ {i.e. as Per time stamp order}

Basic time stamp ordering protocol Conditions:-

old ↑ young ↑

- Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

① When transaction T_1 issue a Read(A) opⁿ

(i)

T_1	T_2
	R(A)
R(A)	

$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is $T_1 \rightarrow T_2$

If transaction T_1 is allowed to perform this R(A) opⁿ then also behaviour of schedule will be Conflict equivalent to Serial Schedule as per time stamp ordering i.e, $T_1 \rightarrow T_2$

∴ There is T_1 is allowed to Perform this R(A) opⁿ

* Reader-Reader will never Create Problem

(ii)

T_1	T_2
	W(A)
R(A)	

Rollback

$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is $T_1 \rightarrow T_2$

If transaction T_1 is allowed to perform this R(A) opⁿ then behaviour of schedule will not be Conflict equivalent to Serial Schedule as per time stamp ordering i.e, $T_1 \rightarrow T_2$

* T_1 is not allowed to perform this R(A) opⁿ and well rollback T_1
* If transaction T_1 issue a R(A) opⁿ and if $WTS(A) > TS(T_1)$, then "Rollback T_1"

Basic time stamp ordering protocol Conditions :-

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

② When transaction T_1 issue a Write(A) opⁿ

(i)

T_1	T_2
	R(A)
W(A)	

Rollback

$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is $T_1 \rightarrow T_2$

If transaction T_1 is allowed to perform this W(A) opⁿ then behaviour of schedule will not be conflict equivalent to serial schedule as per time stamp ordering {i.e. $T_1 \rightarrow T_2$}

∴ Transaction T_1 is not allowed to perform this W(A) opⁿ, and rollback T_1

* If transaction T_1 issued W(A) opⁿ and $RTS(A) > TS(T_1)$, then Rollback T_1

(ii)

T_1	T_2
	W(A)
W(A)	

Rollback

$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is $T_1 \rightarrow T_2$

If transaction T_1 is allowed to perform this W(A) opⁿ then behaviour of schedule will not be conflict equivalent to serial schedule as per time stamp ordering {i.e. $T_1 \rightarrow T_2$}

∴ Transaction T_1 is not allowed to perform this W(A) opⁿ, and rollback T_1

* If transaction T_1 issued W(A) opⁿ and $WTS(A) > TS(T_1)$, then Rollback T_1

Basic time stamp ordering Protocol Condition:-

Let T_1 & T_2 are two transactions s.t. $TS(T_1) < TS(T_2)$

i.e, T_1 is older than T_2

(1) Let T_1 issue Read(A) opn :-

① If $WTS(A) > TS(T_1)$, then Rollback T_1

② otherwise, T_1 is allowed to perform R(A) opn

ie T_1 will perform R(A) opn and set $RTS(A) = Max(RTS(A), TS(T_1))$

(2) Let T_1 issue Write(A) opn :-

① If $RTS(A) > TS(T_1)$, then Rollback T_1

② If $WTS(A) > TS(T_1)$, then also Rollback T_1

(3) otherwise, T_1 will perform its W(A) opn

And set $WTS(A) = TS(T_1)$

Q:- Consider the following schedule

T_1	T_2	T_3
R(A)		
	R(B)	
W(C)		
		R(B)
		R(C)
	W(B)	
		W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) T.S.$(T_1, T_2, T_3) = (30, 10, 20)$

(b) T.S.$(T_1, T_2, T_3) = (30, 20, 10)$

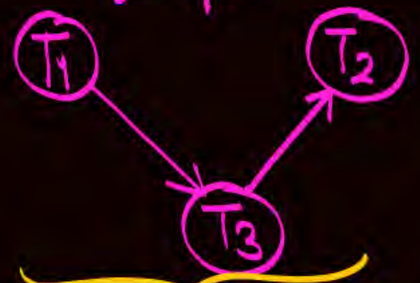
(c) T.S.$(T_1, T_2, T_3) = (10, 20, 30)$

(d) T.S.$(T_1, T_2, T_3) = (10, 30, 20)$

Q:- Consider the following schedule

T_1	T_2	T_3
R(A)		
	R(B)	
		W(C)
		R(B)
		R(C)
	W(B)	
W(A)		

Precedence graph



Acyclic, ∴ CSS

- Conflict equivalent serial schedule is $T_1 \rightarrow T_3 \rightarrow T_2$

For the schedule to be allowed by BTSOP

$TS(T_1) < TS(T_3) < TS(T_2)$ → it is as per "option D"

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) T.S.$(T_1, T_2, T_3) = (30, 10, 20)$

(b) T.S.$(T_1, T_2, T_3) = (30, 20, 10)$

(c) T.S.$(T_1, T_2, T_3) = (10, 20, 30)$

(d) T.S.$(T_1, T_2, T_3) = (10, 30, 20)$

Q:- Consider the following schedule

(a)

30	10	20
T_1	T_2	T_3
R(A)		
	R(B)	
W(C)		
WTS(C)=30		
		R(B)
		R(C) → WTS(C) > TS(T_3) ∴ Rollback T_3 ↓ ∴ Not allowed by BTSOP
	W(B)	
		W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) T.S.(T_1, T_2, T_3) = (30, 10, 20)

(b) T.S.(T_1, T_2, T_3) = (30, 20, 10)

(c) T.S.(T_1, T_2, T_3) = (10, 20, 30)

(d) T.S.(T_1, T_2, T_3) = (10, 30, 20)

Q:- Consider the following schedule

(b)

30 T_1	20 T_2	10 T_3
R(A)		
	R(B)	
W(C) WTS(C)=30		
		R(B)
		R(C) WTS(C)>TS(T_3) ∴ Rollback T_3
	W(B)	
		W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

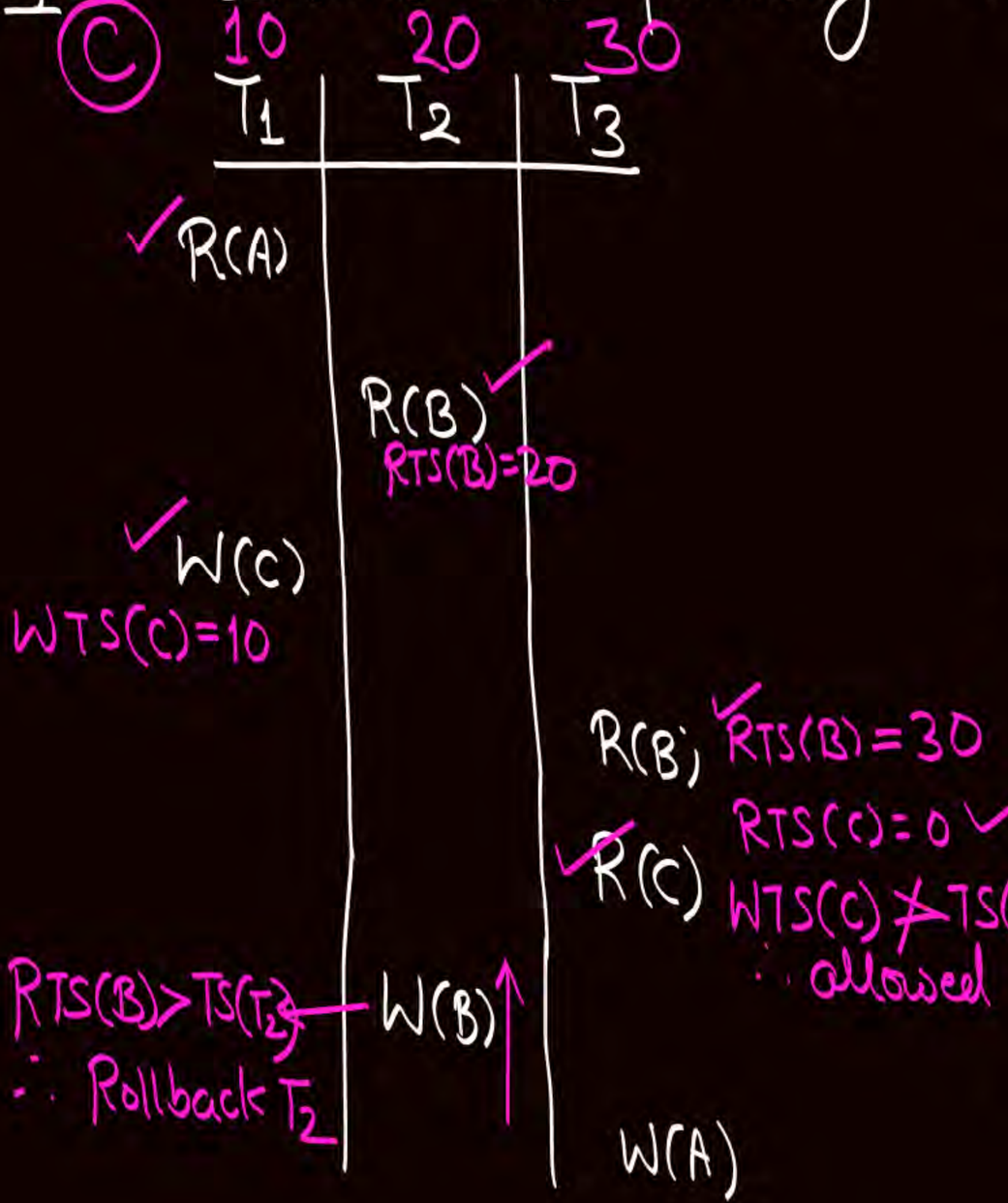
(a) T.S.(T_1, T_2, T_3) = (30, 10, 20)

(b) T.S.(T_1, T_2, T_3) = (30, 20, 10)

(c) T.S.(T_1, T_2, T_3) = (10, 20, 30)

(d) T.S.(T_1, T_2, T_3) = (10, 30, 20)

Q:- Consider the following schedule



Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) T.S.$(T_1, T_2, T_3) = (30, 10, 20)$

(b) T.S.$(T_1, T_2, T_3) = (30, 20, 10)$

(c) T.S.$(T_1, T_2, T_3) = (10, 20, 30)$

(d) T.S.$(T_1, T_2, T_3) = (10, 30, 20)$

Q:- Consider the following schedule

(d)

10	30	20
T_1	T_2	T_3
R(A) ✓		
		R(B) ✓ RTS(B)=30
W(C) ✓ WTS(C)=10		
	R(B) RTS(B)=30	
		R(C) WTS(C) ≯ TS(T_3) ∴ allowed
WTS(B)=0 ✓ RTS(B) ≯ TS(T_2) 30 ≯ 30 ∴ allowed	W(B) ✓	
		WTS(A)=0 ✓ W(A) RTS(A) ≯ TS(T_3) ∴ allowed

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) T.S.(T_1, T_2, T_3) = (30, 10, 20)

(b) T.S.(T_1, T_2, T_3) = (30, 20, 10)

(c) T.S.(T_1, T_2, T_3) = (10, 20, 30)

(d) T.S.(T_1, T_2, T_3) = (10, 30, 20)

Note :-

1. A schedule S is allowed to execute using B.T.S.O.P. if and only if schedule S is a Conflict serializable schedule and Conflict equivalent serial schedule is according to the Time stamp ordering of the transactions
2. If schedule is not a Conflict serializable schedule then it will never be allowed to execute using B.T.S.O.P.
3. If schedule is a C.S.S. but conflict equivalent serial schedule is not as per time stamp ordering of the transactions then schedule is not allowed by B.T.S.O.P.

* B.T.S.O.P Covers only Conflict serializable schedules, but it does not cover the schedules that are View serializable but not Conflict serializable,

∴ We define time stamp ordering protocol with "Thomas write rule"



Topic : Time stamp ordering protocol with Thomas write rule

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

Case ① When transaction T_1 issue a Read(A) oph

(i)

T_1	T_2
	R(A)
R(A)	

(ii)

T_1	T_2
	W(A)
R(A)	

These three case will be exactly same as B.T.S.O.P.

Case ② When transaction T_1 issue a Write(A) oph

(i)

T_1	T_2
	R(A)
W(A)	

(ii)

T_1	T_2
	W(A)
W(A)	

This case will be different from B.T.S.O.P.

Time stamp ordering protocol With Thomas write Rule :-

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

① When transaction T_1 issue a Read(A) oph

(i)

T_1	T_2
	R(A)
R(A)	

$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is
$T_1 \rightarrow T_2$

(ii)

T_1	T_2
	W(A)
R(A)	

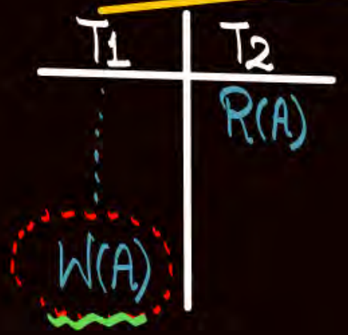
$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is
$T_1 \rightarrow T_2$

Time stamp ordering protocol with Thomas write Rule :-

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

(2) When transaction T_1 issue a Write(A) oph

(i)



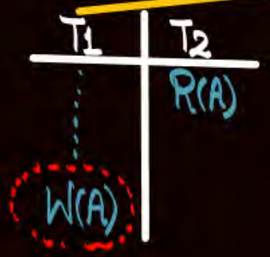
$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is
$T_1 \rightarrow T_2$

Time stamp ordering protocol with Thomas write Rule :-

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

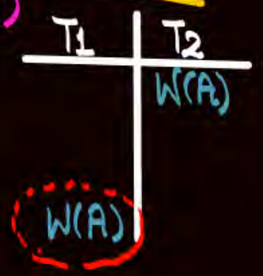
② When transaction T_1 issue a Write(A) opn

(i)



$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is
$T_1 \rightarrow T_2$

(ii)



$TS(T_1) = 10$
$TS(T_2) = 20$
∴ Time stamp ordering is
$T_1 \rightarrow T_2$

Time stamp ordering Protocol with Thomas Write Rule :-

Let T_1 & T_2 are two transactions s.t $TS(T_1) < TS(T_2)$

i.e, T_1 is older than T_2

① Let T_1 issue Read(A) op^n :-

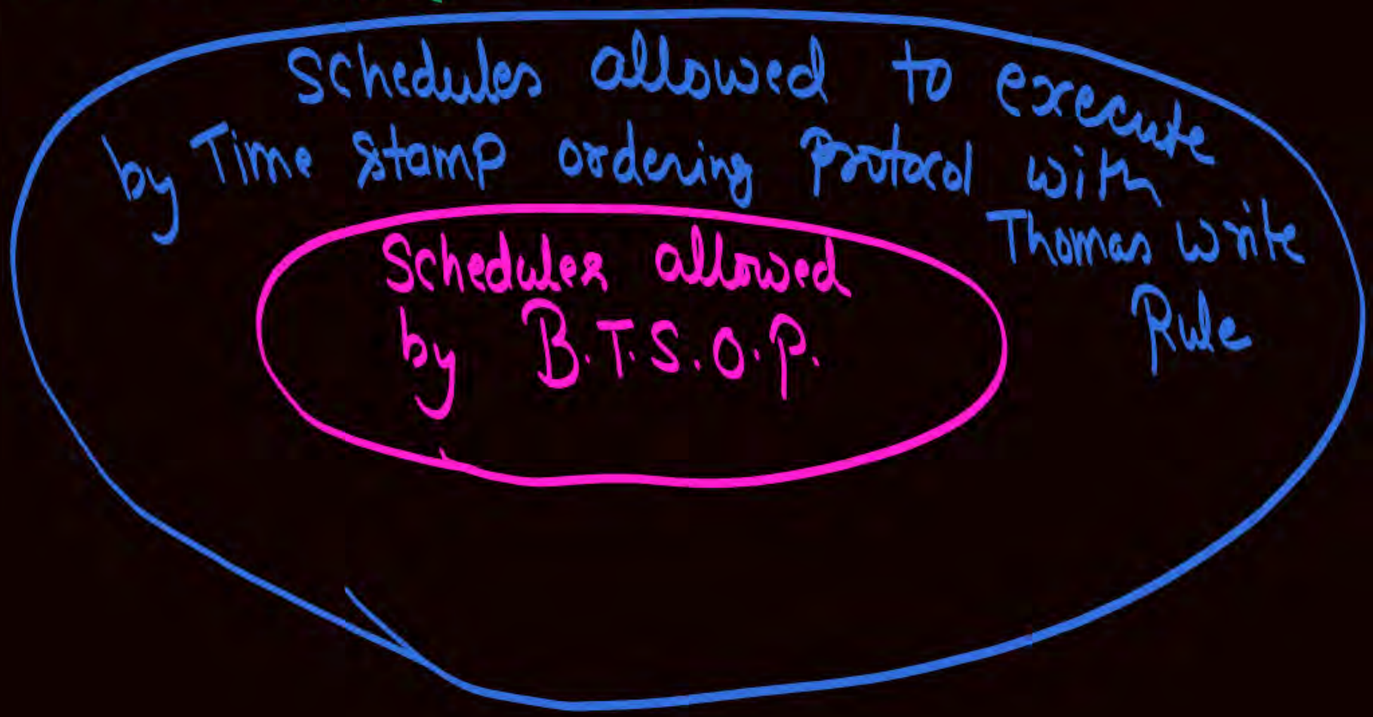
- ⓐ If $WTS(A) > TS(T_1)$, then Rollback T_1
- ⓑ Otherwise, T_1 is allowed to perform the Read(A) op^n
  ∴ T_1 will perform Read(A) op^n.
  And set $RTS(A) = Max(RTS(A), TS(T_1))$

② Let T_1 issue Write(A) op^n :-

- ⓐ If $RTS(A) > TS(T_1)$, then Rollback T_1.
- (or) ⓑ If $WTS(A) > TS(T_1)$, then ignore this w(A) op^n of transaction T_1
  And Continue with the other operations of transaction T_1
- (or) ⓒ Otherwise, I_1 is allowed to perform Write(A) op^n.
  ∴ T_1 will perform the write(A) op^n
  And Set $WTS(A) = TS(T_1)$

Notes :-

① Every schedule which is allowed by B.T.S.O.P. is allowed to execute using Time stamp ordering Protocol with Thomas write Rule, But Converse of the statement need not be true



Note:- (2)

If A schedule is allowed to execute using Time Stamp ordering protocol with Thomas write Rule, then schedule produced by Time Stamp ordering protocol with Thomas write Rule will be view equivalent to the serial schedule based on time stamp ordering of the transactions.

Note :- ③ Both B.T.S.O.P & Time stamp ordering protocol with Thomas write rule are free from deadlock, but starvation is possible

④ Schedules allowed by B.T.S.O.P and/or Time stamp ordering protocol with Thomas write rule may suffer from irrecoverability, cascading rollback and lost-update problem, To solve that problem we can implement Strict time stamp ordering protocols.



Topic : Deadlock prevention algorithms

- We can use the Concept of time stamps assigned by DBMS to prevent the deadlock in lock based Concurrency Control Protocols.



Topic : Deadlock prevention algorithms

There are two algorithms (Protocols) that can be used to prevent the the formation of cycles in the dependency graphs, and hence can be used to prevent the deadlock in lock based protocols.

Those algorithm are

1. Wait - die protocol
2. Wound - wait protocol.

* Using Wait-die & Wound-wait we can Prevent from deadlock in lock based protocols, but Starvation will still be possible.



2 mins Summary

- **Topic** Basic time stamp ordering protocol
- **Topic** Time stamp ordering protocol with Thomas write
- **Topic** Deadlock prevention using time stamp
- **Topic** Wait-Die protocol
- **Topic** Wound-Wait protocol

THANK - YOU

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